

Chapter 1 - Rings and Ideals

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Exercise 1

Suppose $x^r = 0$. Notice

$$(1+x)(1-x+x^2-x^3+\dots+(-1)^{r-1}x^{r-1}) = 1-x^r = 1$$

So $1+x$ is a unit in A . □

Exercise 2

(i) (a) ($\Rightarrow$):

Let $g = b_0 + b_1x + \dots + b_mx^m$ be the inverse of f . Now

$$(a_0 + a_1x + \dots + a_nx^n)(b_0 + b_1x + \dots + b_mx^m) = 1$$

By checking the coefficients of $x^{n+m}, x^{n+m-1}, x^{n+m-2}, \dots$, we get

$$\begin{cases} a_nb_m = 0 \\ a_n^2b_{m-1} = 0 \\ a_n^3b_{m-2} = 0 \\ \vdots \\ a_n^mb_1 = 0 \\ a_n^{m+1}b_0 = 0 \end{cases}$$

On the other hand, by checking the constant term directly,

$$a_0b_0 = 1$$

Therefore a_0 is unit, and $a_n^{m+1} = a_n^{m+1}b_0a_0 = 0$. So a_n is nilpotent.

Moreover, $-a_nx^n$ is a nilpotent element in $A[x]$. By *Ex.1*,

$$f - a_nx^n = a_0 + a_1x + \dots + a_{n-1}x^{n-1}$$

is a unit in $A[x]$, so by iterating the process above we'll see $a_{n-1}, \dots, a_1$ are nilpotent.

(b) ($\Leftarrow$):